

Joshua Gray
Vaijayanti Deshmuk
Lasya Raghavendra

Assignment 9: Disaster Recovery

Step 1: Create EC2

The screenshot displays the AWS Management Console interface for an EC2 instance. The left sidebar shows the navigation menu with options like Dashboard, EC2 Global View, Events, Instances, Images, Elastic Block Store, Network & Security, Load Balancing, Auto Scaling, and Settings. The main content area shows the 'Instance summary for i-03385d71b5de2f2c0 (my-ec2-A9)'. The instance is in the 'Running' state. Key details include: Public IPv4 address: 54.90.149.84, Private IPv4 address: 10.0.1.144, Public DNS: ec2-54-90-149-84.compute-1.amazonaws.com, Private DNS: ip-10-0-1-144.ec2.internal, Instance type: t3.micro, VPC ID: vpc-0f92b839e40c9222e, Subnet ID: subnet-045ddb5a11f2a6af, and Instance ARN: arn:aws:ec2:us-east-1:776781950279:instance/i-03385d71b5de2f2c0. The bottom of the console shows the footer with copyright information and links to CloudShell, Feedback, and Console Mobile App.

Apache was installed at launch using User Data script

```
[ec2-user@ip-10-0-1-144 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Fri 2025-11-07 21:02:53 UTC; 10min ago
     Docs: man:httpd.service(8)
   Main PID: 3190 (httpd)
    Status: "Total requests: 3; Idle/Busy workers 100/0;Requests/sec: 0.00462; Bytes served/sec: 3 B/sec"
      Tasks: 230 (limit: 1053)
     Memory: 16.5M
        CPU: 682ms
    CGroup: /system.slice/httpd.service
            └─ 3190 /usr/sbin/httpd -DFOREGROUND
               3315 /usr/sbin/httpd -DFOREGROUND
               3319 /usr/sbin/httpd -DFOREGROUND
               3320 /usr/sbin/httpd -DFOREGROUND
               3321 /usr/sbin/httpd -DFOREGROUND
              25636 /usr/sbin/httpd -DFOREGROUND

Nov 07 21:02:53 ip-10-0-1-144.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Nov 07 21:02:53 ip-10-0-1-144.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Nov 07 21:02:53 ip-10-0-1-144.ec2.internal httpd[3190]: Server configured, listening on: port 80
[ec2-user@ip-10-0-1-144 ~]$ sudo systemctl status apache2
```

Step 2: Create Snapshot of the volume

The screenshot shows the AWS Management Console interface. At the top, a green notification banner states: "Successfully created snapshot snap-06a4da33e5ee7f379 from volume vol-00aba696adb6b30ee. If you need your snapshot to be immediately available consider using Fast Snapshot Restore." Below this, the "Volumes (1)" section is active, displaying a table with one volume: vol-00aba696adb6b30ee, which is a gp3 type, 8 GiB in size, with 3000 IOPS and 125 Throughput. A "Snapshot summary" box indicates that 0 out of 3 volumes have been backed up. The left sidebar shows the navigation menu with "Instances" and "Images" expanded. The bottom of the console shows the footer with copyright information and links to CloudShell, Feedback, and the Console Mobile App.

Successfully created snapshot snap-06a4da33e5ee7f379 from volume vol-00aba696adb6b30ee. If you need your snapshot to be immediately available consider using Fast Snapshot Restore.

Manage fast snapshot restore

Volumes (1) Info

Saved filter sets

Choose filter set

Search

Volume ID = vol-00aba696adb6b30ee

Clear filters

	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume ID	Create
		vol-00aba696adb6b30ee	gp3	8 GiB	3000	125	snap-051ef35...	-	2025/

Fault tolerance for all volumes in this Region

Snapshot summary

Recently backed up volumes / Total # volumes

0 / 3

Last updated on Fri, Nov 07, 2025, 04:16:29 PM (GMT-05:00)

Data Lifecycle Manager default policy for EBS Snapshots status

No default policy set up | Create policy

CloudShell Feedback Console Mobile App

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Step 3: Create image using snapshot

The screenshot shows the AWS Management Console interface for the "snap-06a4da33e5ee7f379" snapshot. A blue notification banner at the top states: "Successfully requested new image ami-0927fd31d9971adb6. The image is being created. The image-creation process can take several minutes to complete." Below this, the "snap-06a4da33e5ee7f379" details are shown. The "Details" section includes the Snapshot ID (snap-06a4da33e5ee7f379), Owner (776781950279), Description (east-web-root-snap), and Source volume (vol-00aba696adb6b30ee, 8 GiB). The "Full snapshot size" is 1.68 GiB, and it was started on Fri Nov 07 2025 16:17:37 GMT-0500. The "Progress" is 100% and the "Snapshot status" is "Completed". The "Fast snapshot restore" status is also shown. The "Encryption" section indicates that the snapshot is not encrypted. The "Snapshot settings" tab is active, showing the "Snapshot Lock" status as "new". The left sidebar shows the navigation menu with "Instances" and "Images" expanded. The bottom of the console shows the footer with copyright information and links to CloudShell, Feedback, and the Console Mobile App.

Successfully requested new image ami-0927fd31d9971adb6. The image is being created. The image-creation process can take several minutes to complete.

snap-06a4da33e5ee7f379

Last updated less than a minute ago

Delete Actions

Details

Snapshot ID

snap-06a4da33e5ee7f379

Owner

776781950279

Description

east-web-root-snap

Source volume

Volume ID

vol-00aba696adb6b30ee

Volume size

8 GiB

Full snapshot size

1.68 GiB

Started

Fri Nov 07 2025 16:17:37 GMT-0500 (Eastern Standard Time)

Progress

100%

Product codes

-

Snapshot status

Completed

Fast snapshot restore

-

Encryption

Encryption

Not encrypted

KMS key ID

-

KMS key alias

-

KMS key ARN

-

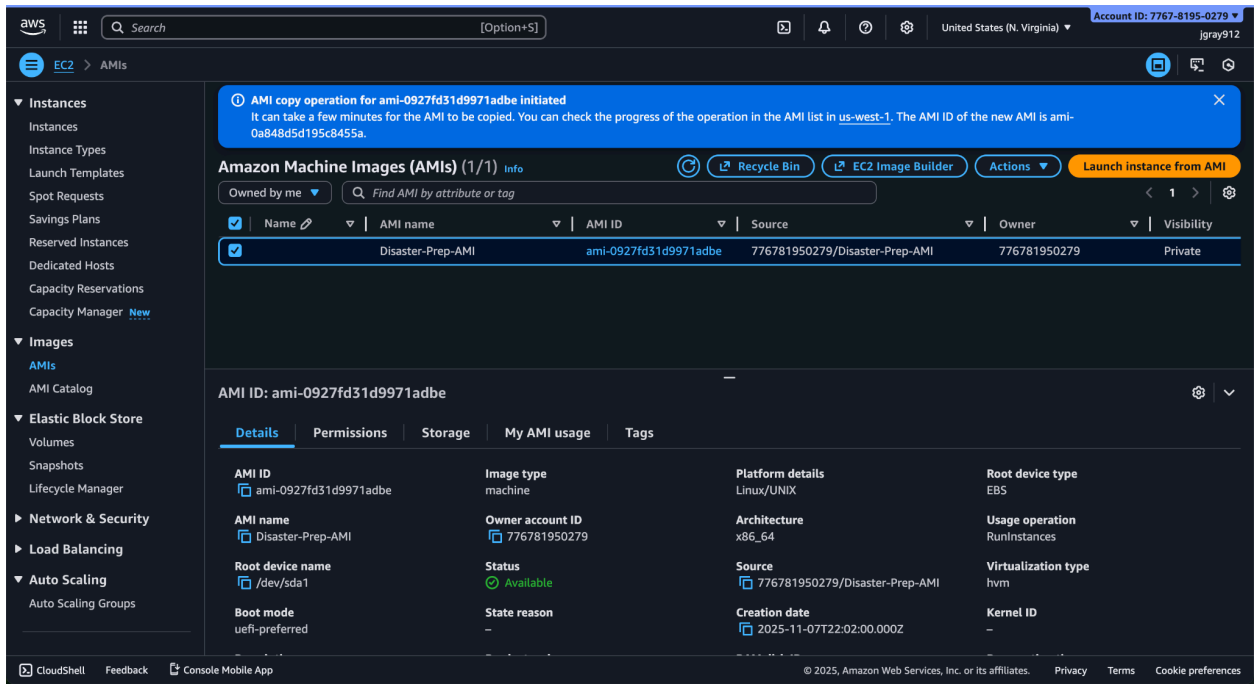
Snapshot settings Storage tier Tags

Snapshot Lock - new

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Step 4: Copy AMI image to us-west region



AMI copy operation for ami-0927fd31d9971adbe initiated
It can take a few minutes for the AMI to be copied. You can check the progress of the operation in the AMI list in [us-west-1](#). The AMI ID of the new AMI is ami-0a848d5d195c8455a.

Amazon Machine Images (AMIs) (1/1) Info

Owned by me Find AMI by attribute or tag

Name	AMI name	AMI ID	Source	Owner	Visibility
☒	Disaster-Prep-AMI	ami-0927fd31d9971adbe	776781950279/Disaster-Prep-AMI	776781950279	Private

AMI ID: ami-0927fd31d9971adbe

Details Permissions Storage My AMI usage Tags

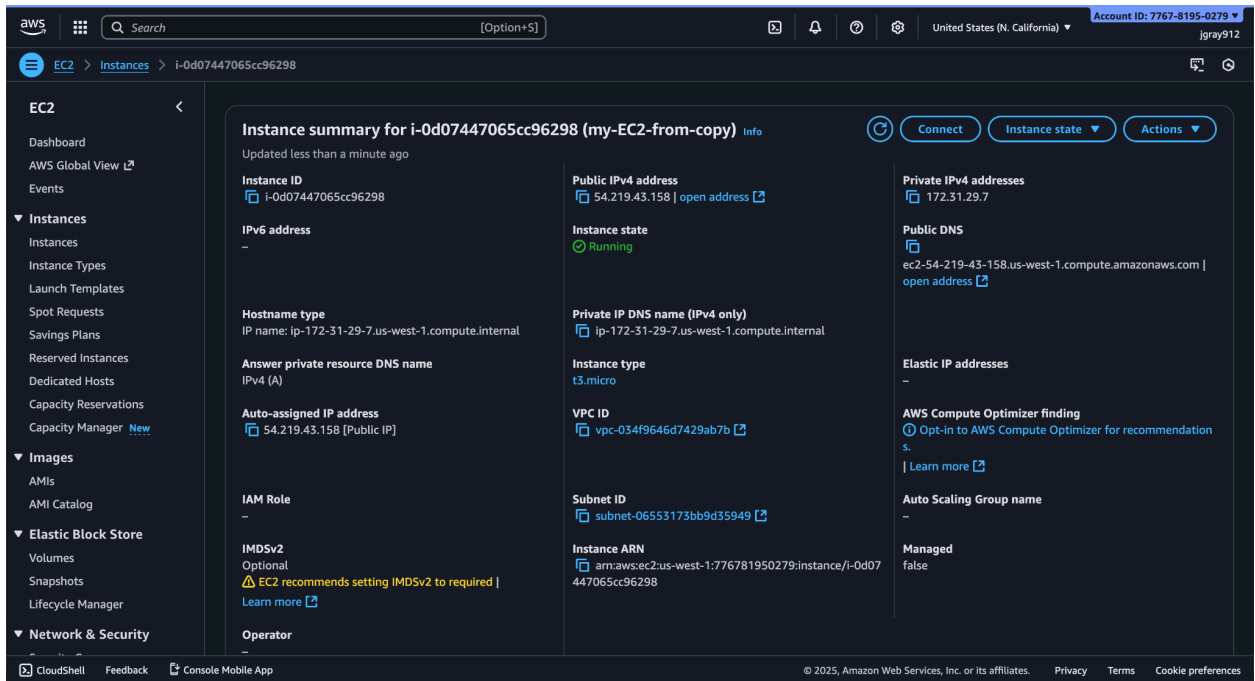
AMI ID	Image type	Platform details	Root device type
ami-0927fd31d9971adbe	machine	Linux/UNIX	EBS

AMI name	Owner account ID	Architecture	Usage operation
Disaster-Prep-AMI	776781950279	x86_64	RunInstances

Root device name	Status	Source	Virtualization type
/dev/sda1	Available	776781950279/Disaster-Prep-AMI	hvm

Boot mode	State reason	Creation date	Kernel ID
uefi-preferred	-	2025-11-07T22:02:00.000Z	-

Step 5: Create EC2 from image in us-west region



EC2 > **Instances** > i-0d07447065cc96298

Instance summary for i-0d07447065cc96298 (my-EC2-from-copy) Info

Updated less than a minute ago

Instance ID
i-0d07447065cc96298

IPv6 address
-

Hostname type
IP name: ip-172-31-29-7.us-west-1.compute.internal

Answer private resource DNS name
IPv4 (A)

Auto-assigned IP address
54.219.43.158 [Public IP]

IAM Role
-

IMDSv2
Optional
EC2 recommends setting IMDSv2 to required | [Learn more](#)

Operator
-

Public IPv4 address
54.219.43.158 | [open address](#)

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-29-7.us-west-1.compute.internal

Instance type
t3.micro

VPC ID
vpc-034f9646d7429ab7b

Subnet ID
subnet-06553173bb9d35949

Instance ARN
arn:aws:ec2:us-west-1:776781950279:instance/i-0d07447065cc96298

Private IPv4 addresses
172.31.29.7

Public DNS
ec2-54-219-43-158.us-west-1.compute.amazonaws.com | [open address](#)

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendation s. | [Learn more](#)

Auto Scaling Group name
-

Managed
false

Step 6: Connect to new instance and access apache

```
(base) joshuagray@MacBook-Pro-6 ~ % chmod 400 Downloads/my-keypair.pem  
joshuagray@MacBook-Pro-6 ~ % ssh -L localhost:8888:localhost:8888 -i Downloads/my-keypair.pem ec2-user@54.219.43.158
```

```
+  
#_____  
~\##### Amazon Linux 2023  
~\#####\  
~\###!  
~\#/ --- https://aws.amazon.com/linux/amazon-linux-2023  
~-V-' '->  
~~~~~  
~/m/'
```

```
Last login: Fri Nov 7 21:11:16 2025 from 50.144.244.225  
[ec2-user@ip-172-31-29-7 ~]$ sudo systemctl status httpd  
● httpd.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)  
   Active: active (running) since Fri 2025-11-07 22:33:26 UTC; 4min 4s ago  
     Docs: man:httpd.service(8)  
 Main PID: 1407 (httpd)  
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"  
      Tasks: 177 (limit: 1053)  
    Memory: 18.5M  
       CPU: 325ms  
   CGroup: /system.slice/httpd.service  
           └─1407 /usr/sbin/httpd -DFOREGROUND  
             └─1479 /usr/sbin/httpd -DFOREGROUND  
               └─1480 /usr/sbin/httpd -DFOREGROUND  
                 └─1482 /usr/sbin/httpd -DFOREGROUND  
                   └─1500 /usr/sbin/httpd -DFOREGROUND
```

```
Nov 07 22:33:25 ip-10-0-1-144.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...  
Nov 07 22:33:26 ip-10-0-1-144.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.  
Nov 07 22:33:26 ip-10-0-1-144.ec2.internal httpd[1407]: Server configured, listening on: port 80  
[ec2-user@ip-172-31-29-7 ~]$
```